

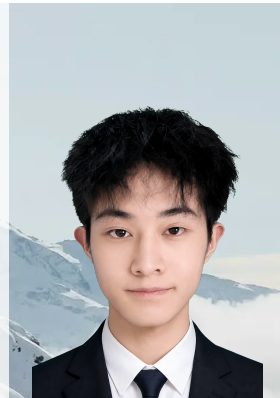
Yongxue Xu (Jerry)

Undergraduate researcher at Sun Yat-sen University

Video generation • World models • 4D scene understanding

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Profile

Undergraduate researcher working on video understanding, object-centric temporal modeling, and 4D dynamic reasoning. Seeking an industry research internship in multimodal foundation models, video generation, or world models.

Available from October 2026 for 6+ months, through graduation in June 2027.

Education & Toolkit

Sun Yat-sen University

SEP 2023 -- JUN 2027

B.Eng. in Intelligent Science and Technology

GPA 3.91 / 4.0 • Major GPA 3.94 / 4.0

Project 985 / 211 / Double First-Class

RESEARCH

Video understanding/generation, multimodal foundation models, world models, 4D scenes, WAM.

TOOLS & LANGUAGES

Python, C/C++, MATLAB, PyTorch; Chinese (native), English (CET-6).

Research Experience

LMM-Track4D: Eliciting 4D Dynamic Reasoning in LMMs via Trajectory-Grounded Dialogue

Co-first author

NeurIPS 2026 submission

Developed [LMM-Track4D](#) for multi-view 4D reasoning with geometry-aware encoding, persistent state memory, and trajectory decoding. Co-developed Track4D-Bench with 526 dialogues, 23.5k frames, and 7.5k object annotations.

Achieved substantial gains over the base model on external benchmarks, particularly in dynamics reasoning, alongside 61.9% trajectory-step accuracy within 0.5 m on Track4D-Bench.

Contribution: Led model design and main experiments; co-developed the benchmark.

Learning to Reason with Persistent Object States for Video Instance Segmentation

Co-first author

ICLR 2027 submission

Developed [POSReasoner](#) for long-horizon identity reasoning, coupling persistent object memory with verified state transitions to counter identity drift and memory corruption under occlusion and reappearance.

Enhanced object recovery on LVOS v1: +13.1 $\mathcal{J}$ after 11–30 absent annotated frames and +3.5 J&F overall, without fine-tuning SAM3.

Contribution: Led method and architecture design, implementation, experiments, and manuscript preparation.

Graph Domain Adaptation Does Not End with Representation Learning

Co-first author

ICLR 2027 submission

Developed [EviGDA](#): independently trained graph-aware and feature-only predictors, entropy-aware domain alignment, and fixed-weight probability fusion without learned routing or target pseudo-labels.

Achieved state-of-the-art average Macro-F1: 67.67 across 16 transfer tasks on 10 datasets (five runs).

Contribution: Designed the core method and led implementation, evaluation, and manuscript development.

Solving the Robust Influence Maximization Problem in Competitive Multilayer Networks via a Diffusion-Aware Role-Guided Evolutionary Approach

First author / TNSE

under review

Extended robust competitive influence maximization (RCIM) from single-layer to multilayer networks with intra-layer competition, cross-layer collaboration, and node-removal robustness. Developed [DRCIM-ML](#), using diffusion-aware role alignment and evolutionary search to transfer seed candidates across layers.

Contribution: Led problem formulation, method design, experiments, and writing; one patent pending.

Selected Publications

LMM-Track4D: Eliciting 4D Dynamic Reasoning in LMMs via Trajectory-Grounded Dialogue 2026

Chaoyue Li*, Yongxue Xu*, Jie Feng*, and Jiayu Ding†

Submitted to NeurIPS 2026; [arXiv:2605.19390](#)

Learning to Reason with Persistent Object States for Video Instance Segmentation 2026

Yongxue Xu*, Boxue Yang*, Ziqian Liu, Shaoqiu Zhang, and Linfeng Zhang†

Submitted to ICLR 2027

Graph Domain Adaptation Does Not End with Representation Learning 2026

Yongxue Xu*, Ziqian Liu*, Enze Zhang, and Maolin Wang†

Submitted to ICLR 2027

Solving the Robust Influence Maximization Problem in Competitive Multilayer Networks via a Diffusion-Aware Role-Guided Evolutionary Approach 2026

Yongxue Xu, Ziqian Liu, Yongqing Huang, and Shuai Wang†

Under review at IEEE Transactions on Network Science and Engineering

Identifying Critical Nodes with Deep Learning and Reinforcement Learning: A Case Study on Urban Road Networks 2025

Zhonghan Liu, Wenshuo Liu, Yongxue Xu, Ziqian Liu, Yichen Liu, and Shuai Wang†

GBCESC 2025 / Best Paper Award

* Equal contribution. † Corresponding author.

Research Initiatives & Collaborations

Collaborate with researchers at **HKUST**, **SJTU**, and **Westlake University** on video generation and multimodal spatiotemporal understanding. Together with peers in **InkMind.AI**, explore multimodal understanding and generation and refine AI-assisted research workflows through hands-on projects.

Additional Research

Urban road-network analysis (GBCESC 2025, Best Paper): contributed dataset construction, GNN/DQN baseline reproduction, and experimental visualization for critical-node identification.

Awards & Competitions

Sun Yat-sen University First-Class Scholarship 2023–2026

Chinese Mathematics Competition (CMC), Guangdong Region – Second Prize Dec. 2024

Mathematical Contest in Modeling (MCM), USA – Honorable Mention Feb. 2025

National Undergraduate Smart Car Competition, South China Region – Second Prize Aug. 2025

Beyond Research

Table tennis, music, and nature travel.